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How can we detect cheating effectively without compromising system security through kernel-level access?

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I thank my friends for introducing me to Counter-Strike.

CONTENTS

List of Figures	iv
List of Tables	v
Introduction	1
0.1 Context	1
0.2 Problematic	2
0.3 Outline	3
1 State of the Art - Cheats and Anti-Cheats	4
1.1 Cheat-Based Exploration of System Architecture	4
1.1.1 Automation Cheats	4
1.1.2 Information Cheats	6
1.1.3 Advanced Cheats	8
1.2 Anti-Cheat Mechanisms	11
1.2.1 Client-Side Anti-Cheats	11
1.2.2 Kernel-Level Anti-Cheats	11
1.2.3 Data-Driven Detection Methods	11
2 Assessing Risks and Limitations of Kernel-Level Anti-Cheats using Windows-Kernel-Explorer	12
2.1 Structural Limitations of Kernel-Level Anti-Cheats	12
2.1.1 Technological Arms Race	12
2.1.2 Limits Against New Forms of Cheating	13
2.1.3 Economic and Operational Costs	14
2.2 Experimental Section: Illustrating Potential Intrusiveness	15
2.2.1 Experimental Objective	15
2.2.2 Experimental Observations	16
2.2.3 Critical Analysis	16

3	Server-Side Approaches: Development of a Gameplay Analytics Pipeline for Suspicion Scoring	17
3.1	Pipeline Implementation	18
3.1.1	Ingestion and Scoring Objectives	18
3.1.2	Algorithmic Architecture and Core Concepts	18
3.1.3	Practical System Deployment	18
3.2	Dataset and Features	18
3.2.1	Telemetry Data Model	18
3.2.2	Data Sources and Profiling	18
3.2.3	Data Constraints and Limitations	19
3.3	Benchmark Execution and Raw Outputs	19
3.3.1	Experimental Protocol	19
3.3.2	Raw Evaluation Results	19
3.4	Analysis and Performance Discussion	19
3.4.1	Net Detection Results	19
3.4.2	Mitigating the High-Skill Bias	19
3.4.3	Future Optimizations	19
3.5	Comparison with Kernel-level approach	20
	Conclusion	21
	Bibliography	22
	Glossary	23
	Acronyms	26
A	Socio-Legal, Privacy, and Community Dynamics of Kernel-Level Anti-Cheat Implementations	27
A.1	System Visibility Level and Perceived Intrusion	27
A.2	Social Perception and Community Acceptance	28
A.3	Legal and Regulatory Questions	29
B	Game Context and Rules	30
B.1	Competitive Format	30
B.2	Core Mechanics Relevant to Cheating	30

C	Why Cheating Breaks Game Balance: The Bullet Penetration	
	Case	32
	C.1 The Penetration Power Mechanic	32
	C.2 Link to Information Cheats	33
D	Client/Server Architecture and Network Communication	34
	D.1 The Authoritative Server Model	34
	D.2 Snapshot Scope and Client Memory	34
	D.3 Network Channel Encryption	34
	D.4 Memory-Based Exploit Surface	35
	D.5 DMA Cheats	35
E	Other appendixes	36
	E.1 Cheaters in CS2	36
	E.2 Existing data aggregation tool	38

LIST OF FIGURES

1	Comparison of the recoil from the AK-47 in-game with its theoretical model, implying that the recoil is deterministic and can be compensated for with a script.	5
2	Comparison of wallhack demonstration using built-in command "r_aoproxy_show 1" in CS2, which forces the engine to render near entity models regardless of occlusion (top) and an ad from a cheating software provider showing wallhack in action (bottom)	7
3	Radarhack demonstration showing the extraction of enemy positions from the game's entity list and their projection onto a secondary interface	8
4	Privilege rings in an OS' structure, highlighting the elevated access of kernel-level cheats in Ring 0 compared to user-mode applications in Ring 3	9
5	Infrastructure of a DMA cheat setup, where a specialized PCIe card reads the host system's RAM directly and transmits the data to an external machine for processing and input spoofing	10
6	User Segmentation & Acceptance Divide, highlighting the contrasting perspectives of competitive and casual players regarding intrusive anti-cheat measures	28
7	Data aggregation made by a CS2 player to show the chance of a cheater being in a game	37
8	Existing data aggregation tool to help visualize a player's game statistics	38

LIST OF TABLES

INTRODUCTION

If you look at a multi-player game, it's the people who are playing the game who are often more valuable than all of the animations and models and game logic that's associated with it.

– GABE NEWELL

0.1 Context

Online competitive video games, particularly first-person shooters (FPS), now occupy a central place in both the video game industry and the esports ecosystem. Their model relies on technical fairness between players: each participant must be subject to the same rules, mechanical constraints, and informational limitations.

This is nowhere more visible than in Counter-Strike 2 (CS2), Valve's flagship tactical shooter and one of the most-watched titles in professional esports. Major tournaments distribute cash prizes reaching several million dollars, and top-tier organizations field full-time professional rosters whose careers – and livelihoods – depend on the perceived integrity of every match. This economic stake is precisely why most high-level CS2 tournaments are played in LAN (Local Area Network) conditions: bringing players into a single physical venue, on hardware controlled by tournament organizers, drastically reduces the attack surface available to cheaters compared to online play, where each participant's machine remains outside the organizers' control. The contrast between these two environments – tightly supervised LAN finals versus the millions of anonymous online matchmaking games played daily – illustrates the scale of the challenge: competitive integrity cannot realistically be guaranteed everywhere through physical supervision alone.

However, cheating represents a growing threat to this balance. The emergence of automated tools such as aimbots and wallhacks alters match integrity, undermines competitive credibility, and directly affects player trust as well as the economic viability of affected titles. In a game where a single well-placed shot can decide a round, and a round can decide a series worth a cash prize, even a small number of undetected cheaters in the matchmaking pool is enough to erode confidence in the entire ranking and reward system – both for the casual player grinding toward a higher rank, and for the aspiring professional whose statistics may be scouted by a team.

To address this issue, many publishers have progressively adopted kernel-level anti-cheat systems.

These solutions operate with very high privileges, allowing them to monitor memory, processes, and certain hardware interactions within the operating system in order to detect manipulation of the game client. While often effective against certain forms of software cheating, they also raise significant concerns regarding cybersecurity, privacy protection, and user acceptability. The permanent presence of a third-party privileged component increases the system's attack surface, introduces a critical point of failure and fuels public debate about the proportionality between technical intrusion and its intended objective.

Beyond pure cybersecurity vulnerabilities, kernel-level solutions introduce profound privacy and regulatory dilemmas. Operating at the highest privilege layer (Ring 0) theoretically grants these software components unhindered visibility over the user's entire system, including personal files, background applications, and non-game-related data. This continuous monitoring mechanism challenges traditional notions of informed user consent and clashes directly with modern data protection frameworks, such as the GDPR, which mandate strict principles of data minimization and proportionality. Consequently, this technological choice has sparked a sharp divide within the gaming community: while competitive players often tolerate intrusive measures in the name of competitive integrity, casual users increasingly reject what they perceive as a form of digital surveillance, ultimately threatening game adoption and brand reputation.

0.2 Problematic

In this context, the following question arises: can cheating be effectively detected without compromising system security through kernel-level access? This thesis explores an alternative approach primarily based on server-side behavioral analysis and machine learning techniques. Instead of directly inspecting the player's machine, the idea is to leverage gameplay data itself—match statistics, action sequences, and temporal and mechanical consistency—to identify anomalies characteristic of assisted behavior.

Recent research suggests that analyzing replays and gameplay data in games such as Counter-Strike 2 makes it possible to distinguish legitimate players from suspicious profiles using labeled datasets. However, these approaches also present limitations: the need for large volumes of data, the risk of false positives, delayed detection, and the continuous adaptation of cheaters. The objective of this thesis is therefore twofold. First, to analyze the advantages and limitations of kernel-level anti-cheats in light of cybersecurity, system stability, and user trust considerations. Second, to design and evaluate a behavioral detection system based on gameplay metrics. A proof of concept will be implemented using replay data or simulated logs for a Counter-Strike-like FPS in order to generate a suspicion score and study the prioritization of human review.

Finally, the thesis will propose an architectural positioning of these approaches within a multi-layer anti-cheat system, combining server-side detection, lightweight client mechanisms, and human validation, with the aim of reconciling effectiveness, security, and player acceptability.

0.3 Outline

In the first chapter, we will present the state of the art in cheating and anti-cheat technologies. We will analyze the main categories of cheats — automation tools, information cheats, and advanced hardware-assisted vectors — and examine the anti-cheat mechanisms developed in response, from client-side solutions to kernel-level drivers and data-driven detection methods.

In the second chapter, we will assess the risks and structural limitations of kernel-level anti-cheats. We will analyze the technological arms race they sustain, their inability to address hardware-assisted cheating such as DMA, and their economic and operational costs. This analysis will be supported by an experimental section using Windows-Kernel-Explorer to illustrate the level of system access a privileged kernel component holds, and to discuss the security and privacy implications that follow.

In the third chapter, we will present the design and implementation of a server-side gameplay analytics pipeline for suspicion scoring. We will describe the telemetry data model, the algorithmic architecture combining rule-based heuristics and an LSTM layer, and the results of benchmarking against simulated player profiles. We will evaluate the system's performance, discuss the high-skill bias problem and its mitigations, and compare this approach against the kernel-level paradigm examined in the previous chapter.

Finally, in the conclusion, we will synthesize our findings and propose a positioning of these complementary approaches within a multi-layered anti-cheat architecture, balancing detection effectiveness, system security, and player trust.

STATE OF THE ART - CHEATS AND ANTI-CHEATS

Technology is just a tool. In terms of getting the kids working together and motivating them, the teacher is the most important.

– BILL GATES

1.1 Cheat-Based Exploration of System Architecture	4
1.1.1 Automation Cheats	4
1.1.2 Information Cheats	6
1.1.3 Advanced Cheats	8
1.2 Anti-Cheat Mechanisms	11
1.2.1 Client-Side Anti-Cheats	11
1.2.2 Kernel-Level Anti-Cheats	11
1.2.3 Data-Driven Detection Methods	11

1.1 Cheat-Based Exploration of System Architecture

1.1.1 Automation Cheats

Automation cheats represent a critical class of malicious software engineered to augment or entirely substitute a player’s mechanical execution within the competitive environment. This category primarily encompasses automated aiming tools (aimbots), automatic firing utilities (triggerbots), and execution macros or scripts. By targeting the core pillars of first-person shooter (FPS) gameplay—specifically mechanical precision, muscle memory, and cognitive reaction time, these tools disrupt competitive balance by replacing human motor control with software-driven operations.

To achieve this automation, the software must interface directly with either the operating system’s input layer or manipulate the inner variables governing the game engine logic. The depth of this interaction generally defines the cheat’s sophistication and detection difficulty:

- **Input Layer Manipulation:** Client-side automation tools often operate by injecting synthetic mouse and keyboard events directly into the operating system's input stream. A typical trigger-bot, for example, continuously polls the game client's state or memory addresses to determine when an enemy entity's hitbox intersects with the exact coordinates of the player's crosshair. The moment this condition is validated, the cheat programmatically fires a click event into the input queue, completely bypassing human visual processing and nervous system latency. The latency between the detection of the event and the actual input can be configured to try and pass as a human reaction, though the main point of this process is still to be way faster than the average player.
- **Subversion of Game Engine Logic:** More advanced automation mechanisms bypass peripheral input emulation altogether, executing modifications directly within the local game engine client memory space. An aimbot reads the real-time spatial positions (three-dimensional coordinates) of all active opponents from the engine's local entity list. It then calculates the precise angular vector offsets required to bridge the gap between the current viewpoint and the target's head hitbox. By overwriting the client's view-angle variables within the engine memory before the next frame rendering or tick packet serialization, the cheat forces perfect accuracy.

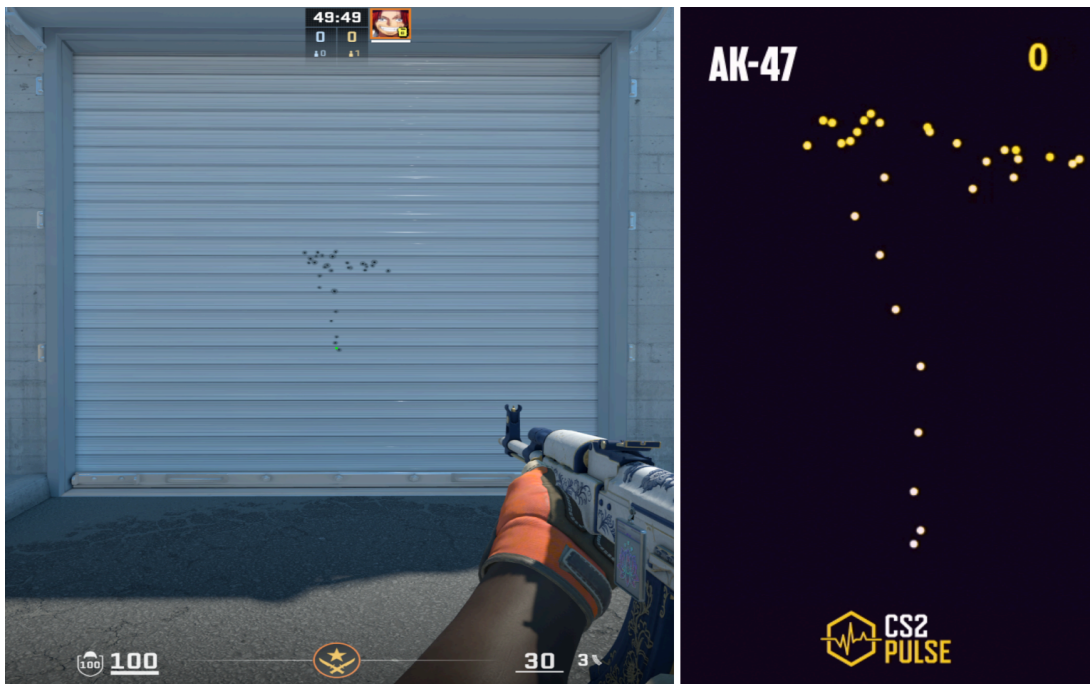


Figure 1: Comparison of the recoil from the AK-47 in-game with its theoretical model, implying that the recoil is deterministic and can be compensated for with a script.

1

This enables frame-perfect recoil compensation and automated target tracking that completely overrides the physical constraints designed by the game developers, as illustrated in Figure 1, which

¹<https://cs2pulse.com/spray-patterns/ak-47/>

confirms the deterministic nature of the AK-47's spray pattern and the trivial feasibility of scripted compensation.

1.1.2 Information Cheats

Unlike automation tools that directly override a player's physical inputs, information cheats focus on subverting the tactical integrity of tactical shooter ecosystems by breaking down spatial boundaries. Tactical first-person shooters heavily rely on a conceptual "fog of war"—a structural design where information regarding enemy positions, utility placement, and movement is deliberately occluded by spatial geometry to reward map control and sensory awareness. Information cheats, primarily manifested as wallhacks and radarhacks, systematically dismantle this baseline restriction. They do not execute actions for the user but instead grant an absolute cognitive advantage by exposing hidden game data and altering the rendering pipeline.

To achieve this asymmetric visualization, these tools typically exploit two distinct technical vectors within the local client system:

- **Rendering Pipeline Exploitation:** Wallhacks operate by intercepting or altering the graphics rendering pipeline responsible for drawing the game world onto the player's screen. In a legitimate game loop, the engine utilizes occlusion to ensure that player models hidden behind opaque geometry (such as walls or solid terrain) are not drawn on the screen. Wallhacks circumvent this by hooking into the graphic APIs used by the engine. By injecting code that forces the graphics card to ignore occlusion parameters when processing enemy entity models, the cheat renders these entities on top of intervening geometry. This is frequently accompanied by ESP (Extra Sensory Perception) boxes, which overlay real-time lines, health bars, and item descriptions directly onto the player's view canvas — a result visually indistinguishable from the engine's own debug rendering mode, as shown in Figure 2, where the output of a native CS2 command forcing occlusion bypass (top) is compared side-by-side with a commercial cheat advertisement (bottom).

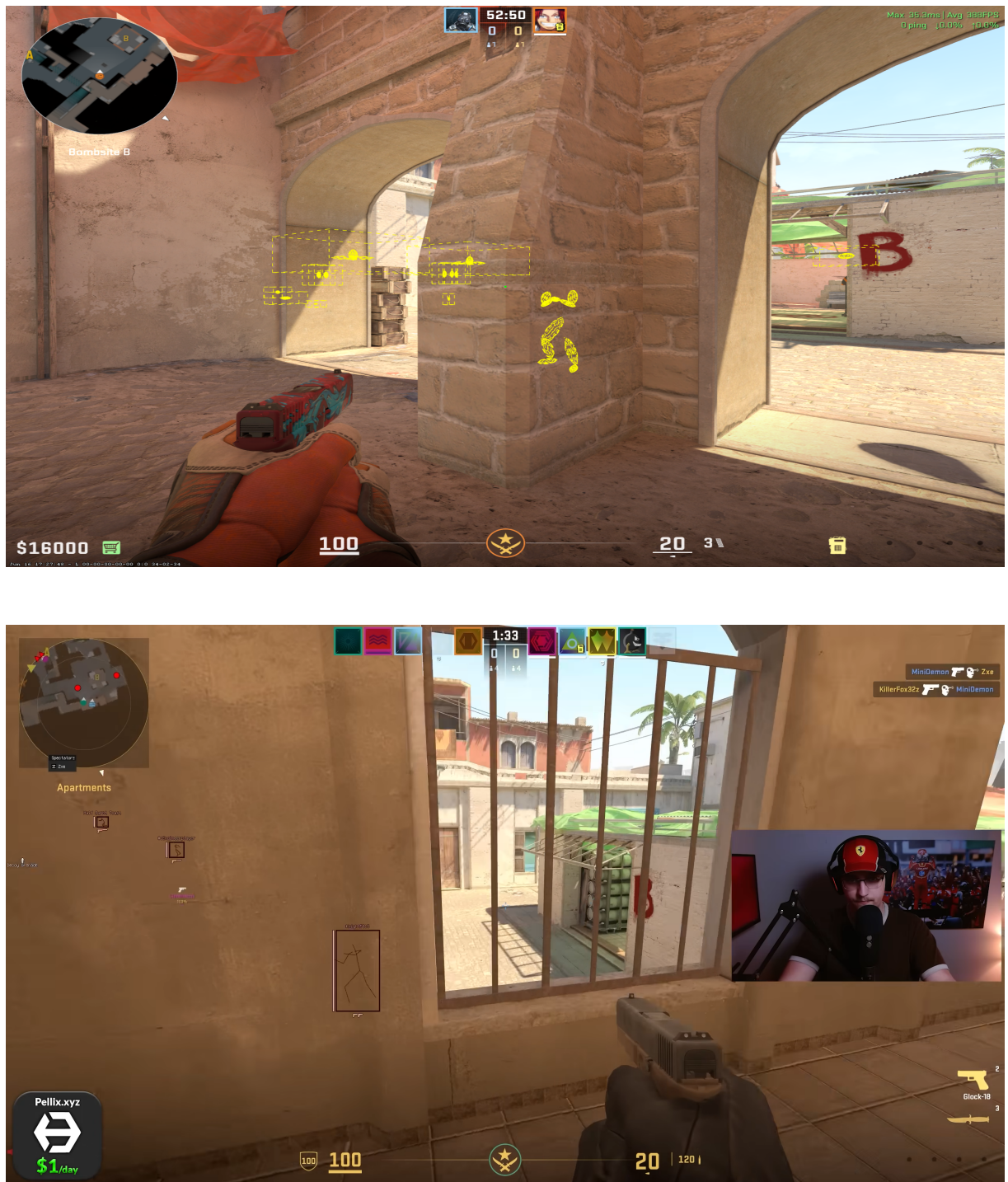


Figure 2: Comparison of wallhack demonstration using built-in command "r_aoproxy_show 1" in CS2, which forces the engine to render near entity models regardless of occlusion (top) and an ad from a cheating software provider showing wallhack in action (bottom)

2

- Game Data Exposure: Radarhacks bypass the visual canvas entirely, targeting the underlying spatial data structures residing within the client's volatile memory or network socket streams.

²<https://www.youtube.com/watch?v=FRmhe9mUGmY>

Every online game client maintains a local "Entity List" containing the precise three-dimensional coordinate vectors of all active participants, transmitted by the server to ensure local interpolation and hit registration. A radarhack continuously scans the process memory space to extract these spatial vectors or sniffs incoming network packets before they are parsed by the client. This data is then projected onto a secondary web-based interface, an external execution window, or a simulated 2D mini-map overlay. Because this methodology leaves the primary visual interface completely clean and does not alter rendering hooks, it provides the cheater with absolute situational awareness while significantly minimizing visibility to automated anti-cheat screen capture mechanisms or human review systems. This type of cheating is one of the hardest to detect as, except for the direct scanning of the process memory space, the behaviour of such cheaters can often be associated with a good game-sense or instincts, hence being harder to prove as blatant cheating — Figure 3 illustrates this extraction pipeline, showing enemy coordinates pulled from the entity list and projected onto a clean secondary interface that leaves the primary game view entirely unaltered.

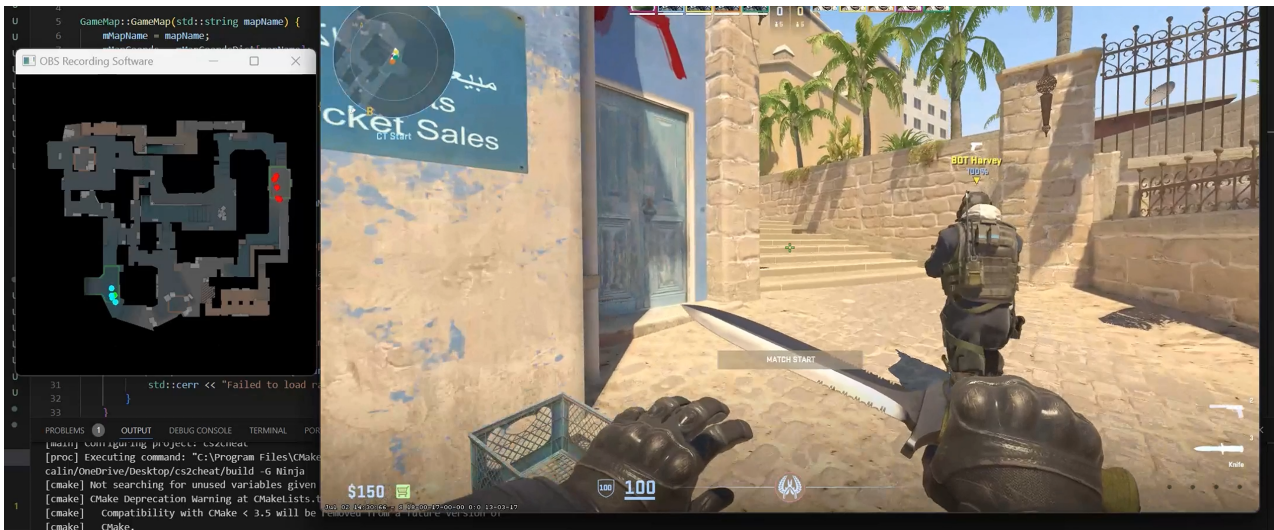


Figure 3: Radarhack demonstration showing the extraction of enemy positions from the game's entity list and their projection onto a secondary interface

3

1.1.3 Advanced Cheats

As client-side anti-cheat mechanisms evolved to actively monitor user-mode operations, the cheating ecosystem shifted toward highly sophisticated execution paradigms categorized as advanced cheats. These advanced threats deliberately bypass the operating system's standard application boundaries by executing within highly privileged environments or utilizing standalone external hardware vectors. By exploiting low-level OS/Kernel interactions and dedicated hardware interfaces, these tools minimize or

³<https://github.com/bordeanu05/cs2cheat>

entirely eliminate their local software footprint, rendering conventional user-mode memory scanners and API hooks ineffective.

- **Kernel-Level Cheats and Kernel interaction:** Kernel-level cheats operate inside the most privileged layer of the computer's processor architecture, known as Kernel Mode (Ring 0). While regular video games and applications are restricted to User Mode (Ring 3) where they have limited permissions, software running in Ring 0 has absolute control over the entire system's memory and hardware. Because modern client-side anti-cheat systems also run at this kernel level to monitor the game, cheat developers must find ways to gain equal authority to hide their presence and subvert security controls — a dynamic directly reflected in the privilege ring architecture shown in Figure 4, where both kernel-level cheats and anti-cheats compete for the same Ring 0 authority, leaving user-mode applications in Ring 3 with no visibility over either.

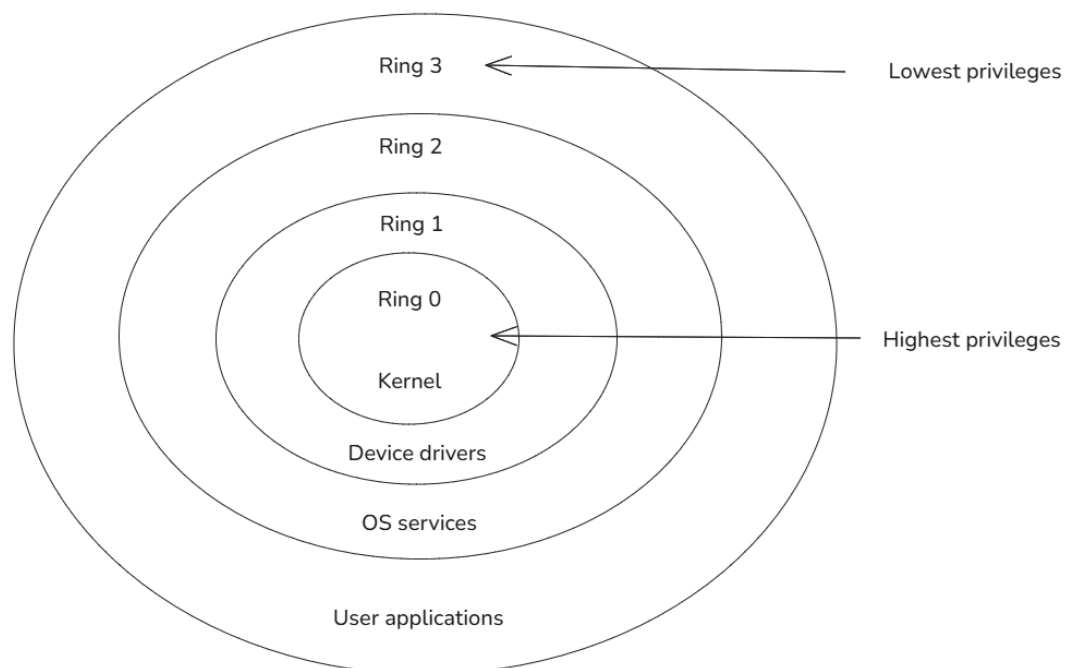


Figure 4: Privilege rings in an OS' structure, highlighting the elevated access of kernel-level cheats in Ring 0 compared to user-mode applications in Ring 3

4

This advanced interaction with the operating system occurs through three primary mechanisms:

- **System Penetration via BYOVD:** Modern operating systems like Windows require all kernel-level software to be digitally signed and approved by Microsoft. To bypass this security check, cheat developers use a technique called Bring Your Own Vulnerable Driver (BYOVD). They take an old, legitimately signed utility driver (from a trusted company) that contains a known security flaw. The cheat loads this trusted driver into the system and exploits its vulnerability to inject unsigned, malicious cheat code directly into the secure kernel memory space.

⁴<https://www.tutorialspoint.com/article/protection-ring>

- Direct System Manipulation (DKOM): Once established inside Ring 0, the cheat can modify the operating system’s internal tracking structures. Using a method known as Direct Kernel Object Manipulation (DKOM), the cheat can erase its own program from the active process list or alter the system’s memory maps (page tables). This disguises the cheat’s memory footprint, making it look like a harmless, native operating system component.
- External Cheats and Hardware Interfaces: The most modern escalation in this technical conflict involves external and hardware-assisted cheats, which decouple the cheat execution environment from the host operating system running the game client. This paradigm relies on physical hardware interfaces, most notably Direct Memory Access (DMA) via specialized hardware expansion cards installed directly into the host machine’s PCI Express (PCIe) slots. The DMA card physically reads and writes to the system’s volatile memory (RAM) directly through the hardware controller, completely bypassing the CPU and any software-level security controls enforced by the operating system kernel. This raw memory stream is transmitted via a physical interface to a secondary, completely isolated computer. This external machine parses the game state asynchronously, calculates visual overlays or aim vectors, and can send synthetic peripheral inputs back to the host via specialized hardware input spoofers or modified mouse firmware. Because no malicious code executes, modifies memory, or hooks processes on the primary gaming machine, hardware-assisted vectors leave zero software footprint for traditional detection engines to scan — Figure 5 details this infrastructure, showing how the PCIe card reads host RAM through the hardware controller and relays the raw memory stream to an isolated external machine responsible for processing and input spoofing.

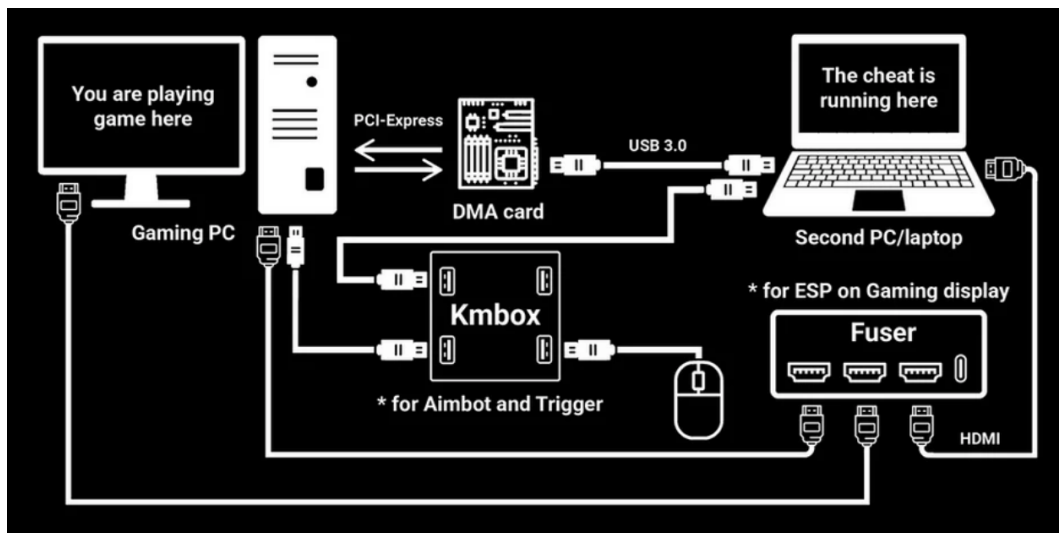


Figure 5: Infrastructure of a DMA cheat setup, where a specialized PCIe card reads the host system’s RAM directly and transmits the data to an external machine for processing and input spoofing

1.2 Anti-Cheat Mechanisms

1.2.1 Client-Side Anti-Cheats

- Code injection detection
- Memory scanning

1.2.2 Kernel-Level Anti-Cheats

- Privileged monitoring
- Detection of low-level cheats

1.2.3 Data-Driven Detection Methods

▶ Shifting the trust anchor to the server

- Reduced dependence on the client
- Global view of game statistics

▶ Player behavior modeling

- Performance patterns
- Cross-match consistency
- Statistical anomalies

▶ Conceptual limitations

- False positives
- Skilled player vs cheater bias
- Exotic but legitimate behaviors

ASSESSING RISKS AND LIMITATIONS OF KERNEL-LEVEL ANTI-CHEATS USING WINDOWS-KERNEL-EXPLORER

The programmers of tomorrow are the wizards of the future. You're going to look like you have magic powers compared to everybody else.

– GABE NEWELL

2.1 Structural Limitations of Kernel-Level Anti-Cheats	12
2.1.1 Technological Arms Race	12
2.1.2 Limits Against New Forms of Cheating	13
2.1.3 Economic and Operational Costs	14
2.2 Experimental Section: Illustrating Potential Intrusiveness . . .	15
2.2.1 Experimental Objective	15
2.2.2 Experimental Observations	16
2.2.3 Critical Analysis	16

2.1 Structural Limitations of Kernel-Level Anti-Cheats

2.1.1 Technological Arms Race

The deployment of kernel-level anti-cheat systems did not resolve the cheating problem — it escalated it. By elevating the detection layer to Ring 0, publishers effectively forced cheat developers to match that privilege level, triggering a continuous cycle of technical escalation in which each defensive advancement is systematically studied, reverse-engineered, and circumvented.

This dynamic is structurally inherent to the kernel-level paradigm. When an anti-cheat driver is deployed, it establishes a set of monitored invariants: a list of known cheat signatures, a set of

protected memory regions, a collection of hooked system calls, and a series of integrity checks targeting the game process. Each of these controls is static at the moment of deployment. Cheat developers, operating in a well-funded and highly motivated ecosystem, treat each new anti-cheat version as a reverse-engineering target. By analyzing the driver's behavior in controlled environments — often through the same kernel-level tooling described in Section 1.1.3 — they systematically map the detection surface and identify evasion vectors before the anti-cheat can update its signatures.

The BYOVD technique described in Section 1.1.3 is a direct product of this arms race: it emerged precisely because kernel-level anti-cheats began enforcing driver signature validation as a first-line defense, which cheat developers countered by exploiting the pre-existing trust chain of legitimately signed third-party drivers. Anti-cheat vendors responded by maintaining and enforcing blocklists of known-vulnerable drivers — a measure that Microsoft has since integrated at the OS level through its Vulnerable Driver Blocklist. Cheat developers, in turn, shifted toward newer or less-documented vulnerable drivers that had not yet been catalogued. Each iteration of this cycle demands a new detection update, a new blocklist entry, and a new engineering investment from the anti-cheat team, while the cheat ecosystem distributes the cost of evasion research across a community of actors sharing tools and findings openly.

This asymmetry in update cadence represents one of the core structural weaknesses of the kernel-level approach. An anti-cheat update is a significant engineering and validation effort: kernel drivers must pass rigorous stability testing before deployment, as a defective kernel component can trigger system-wide crashes (Blue Screens of Death), corrupt system state, or introduce new exploitable vulnerabilities. The cheat ecosystem operates under no such constraint. A new evasion technique can be packaged and distributed within days of an anti-cheat update, while the corresponding detection patch may take weeks or months to safely develop, test, and deploy. The defender is therefore structurally slower than the attacker, regardless of the engineering resources invested.

The consequence is a perpetual dependency on continuous updates that carries compounding costs. Each new game release, operating system update, or hardware generation requires the anti-cheat driver to be adapted and re-validated. A kernel component that is not actively maintained rapidly becomes a liability rather than a protection: its signatures grow stale, its hooks are bypassed by updated cheat loaders, and its continued Ring 0 presence introduces security risk without providing a corresponding detection benefit.

2.1.2 Limits Against New Forms of Cheating

Beyond the arms race dynamic, kernel-level anti-cheats are architecturally incapable of addressing an entire class of modern cheating vectors: those that externalize cheat execution outside the host operating system entirely. This limitation is not a matter of implementation quality or update frequency — it is a fundamental consequence of the threat model kernel-level systems were designed to address.

Kernel-level anti-cheats operate on a core assumption: that the cheat software leaves a detectable footprint within the host OS, whether in the form of unauthorized memory modifications, suspicious process activity, unsigned driver loading, or anomalous system call patterns. This assumption held when the dominant cheat paradigm consisted of injected DLLs and user-mode memory scanners. It breaks down entirely against hardware-assisted cheating and, specifically, against Direct Memory Ac-

cess (DMA) cheats as described in Section 1.1.3. and in Figure 5.

Recall that a DMA cheat operates by connecting a specialized PCIe card directly to the host machine's memory bus. This card reads the game's process memory — including the decrypted Entity List containing all player positions — at the hardware level, bypassing the CPU and the operating system's process model entirely. The extracted game state is transmitted to a physically separate machine, which handles overlay rendering and aim vector computation, and synthetic inputs are fed back to the host via hardware input spoofers. Because no cheat software executes on the host OS, no unsigned driver is loaded, no process is injected, and no memory region on the primary machine is modified by the cheat, the kernel-level anti-cheat has no local artifact to detect. Its monitoring perimeter — however deep into Ring 0 it extends — simply does not reach the hardware interface through which the DMA card operates.

This represents a categorical defeat for the kernel-level paradigm against a class of cheats that is both commercially available and increasingly prevalent in high-stakes matchmaking environments. The hardware required for a functional DMA setup — a PCIe card, a secondary machine, and a compatible input spoofer — is available for purchase at a few hundred dollars, a price point well within reach of motivated cheaters in games where ranked progression carries economic or reputational value.

Efforts to detect DMA cheats at the software level have explored indirect approaches: monitoring PCIe device enumeration for unexpected hardware, measuring anomalous memory access latency patterns, or detecting the firmware signatures of known DMA cards. Each of these mitigations, however, suffers from the same fundamental constraint as the broader arms race: they are signature-based, enumerable, and bypassable. DMA card firmware can be reprogrammed to spoof legitimate device identifiers. Latency patterns can be masked. The attacker controls the hardware and can modify it to defeat any software-level heuristic the anti-cheat deploys.

The architectural conclusion is significant: the kernel-level approach, by design, can only monitor what occurs within the software stack of the host OS. Any cheat vector that operates below that boundary — whether through hardware memory access, external machine processing, or firmware-level input injection — falls entirely outside the detection perimeter of even the most privileged kernel driver.

2.1.3 Economic and Operational Costs

The structural limitations described above do not manifest in a vacuum — they translate into concrete and ongoing economic costs for publishers who choose to deploy kernel-level anti-cheat systems. These costs operate across three distinct dimensions that collectively challenge the long-term viability of the approach.

The first is driver maintenance. A kernel driver is not a product that can be shipped and forgotten: it must be continuously adapted to new OS versions, new hardware generations, and new security policies enforced by Microsoft's evolving Secure Boot and driver signing infrastructure. Each Windows update cycle introduces potential compatibility breaks that must be diagnosed and patched before they manifest as crashes or silent detection failures in the player base. This maintenance burden requires a dedicated engineering team with specialized kernel-level expertise — a skillset that is both scarce and expensive in the current software labor market.

The second is security incident management. As discussed in Appendix A.3, the mandatory installation of a Ring 0 component exposes the publisher to significant liability in the event of a security flaw. The historical record is unambiguous: the Capcom anti-cheat driver exploit allowed arbitrary unsigned code execution at kernel level on every machine where it was installed; the Genshin Impact anti-cheat driver was weaponized by a third party to deploy ransomware by exploiting its legitimate Ring 0 access. Each such incident triggers a crisis response cycle — emergency patching, public disclosure management, potential regulatory scrutiny, and, in severe cases, class-action litigation. The cost of a single major incident can exceed the cumulative engineering investment in the anti-cheat system itself.

The third is reputational and commercial impact. As detailed in Appendix A.2, a growing segment of the player base actively rejects kernel-level solutions, translating publisher anti-cheat decisions into measurable adoption losses. The exclusion of Linux and Steam Deck users — a platform whose market share has grown substantially with the mainstream success of Valve’s handheld — from titles enforcing kernel drivers represents a quantifiable reduction in the addressable player pool. For free-to-play titles where player count directly drives monetization through network effects and cosmetic economies, this trade-off carries direct financial consequences that compound over time.

Taken together, these three cost vectors — maintenance, incident response, and adoption friction — establish that kernel-level anti-cheat is not simply a technical solution with technical limitations, but a strategic commitment with sustained organizational costs. The central question for publishers is therefore whether the detection effectiveness gained justifies this investment, particularly given the categorical limitations against hardware-assisted cheating vectors identified in Section 2.1.2. It is precisely this question that motivates the server-side behavioral approach explored in Chapter 3.

2.2 Experimental Section: Illustrating Potential Intrusiveness

2.2.1 Experimental Objective

- Illustrate the level of access of a privileged component
- Demonstrate security and privacy implications
- Headstart thanks to <https://github.com/AxtMueller/Windows-Kernel-Explorer> -> PROBLEM: ONLY SUPPORTED UNTIL WINDOWS 10. WHAT DO DO INSTEAD? I SEARCHED FOR ALTERNATIVES (READ RAM CONTENT ON LINUX, OTHER TOOLS TO READ RAM CONTENT ON WINDOWS) BUT COULDN'T GET ANYTHING TO WORK ON LINUX NOR WINDOWS. I COULD SHOW HOW CHEAT ENGINE CAN FIND A VARIABLE LOCATION IN RAM FROM USING SUCCESSIVE VALUES TO PINPOINT THE EXACT LOCATION OF THE VARIABLE IN RAM (MEANING YOU CAN THEN READ IT OR MODIFY IT) BUT WOULD THAT BE REALLY USEFULL IN THE CONTENT OF THIS THESIS?

2.2.2 Experimental Observations

- Access to sensitive system resources
- Privilege differences compared to regular software
- Observation of non-game-related data

2.2.3 Critical Analysis

- Real vs theoretical implications
- Link with user perception
- Discussion of effectiveness vs intrusiveness proportionality

SERVER-SIDE APPROACHES: DEVELOPMENT OF A GAMEPLAY ANALYTICS PIPELINE FOR SUSPICION SCORING

We can't solve today's problems with the mentality that created them.

– ALBERT EINSTEIN

3.1 Pipeline Implementation	18
3.1.1 Ingestion and Scoring Objectives	18
3.1.2 Algorithmic Architecture and Core Concepts	18
3.1.3 Practical System Deployment	18
3.2 Dataset and Features	18
3.2.1 Telemetry Data Model	18
3.2.2 Data Sources and Profiling	18
3.2.3 Data Constraints and Limitations	19
3.3 Benchmark Execution and Raw Outputs	19
3.3.1 Experimental Protocol	19
3.3.2 Raw Evaluation Results	19
3.4 Analysis and Performance Discussion	19
3.4.1 Net Detection Results	19
3.4.2 Mitigating the High-Skill Bias	19
3.4.3 Future Optimizations	19
3.5 Comparison with Kernel-level approach	20

3.1 Pipeline Implementation

3.1.1 Ingestion and Scoring Objectives

- Automating telemetry-driven suspicion scoring
- Prioritization matrix for human review (Overwatch-style triage)

3.1.2 Algorithmic Architecture and Core Concepts

- Rule-Based Foundations: Establishing game engine limits, behavioral baselines, and mechanical consistency boundaries
- Hybrid Modeling Framework: Combining a classical deterministic heuristic algorithm (live run execution) with sequential Machine Learning concepts (Long Short-Term Memory - LSTM)
- Flag Weighting and Temporal Aggregation: Make individual mechanical anomalies accumulate into a player history score

3.1.3 Practical System Deployment

- Architecture of the analytics pipeline: <https://github.com/selligre/thesis-cs2-ac-server> (could use <https://github.com/pnxenopoulos/awpy> to help parsing data from demos)
- Event-driven design: Comparison of real-time telemetry parsing vs. asynchronous post-match processing

3.2 Dataset and Features

3.2.1 Telemetry Data Model

- Aim Vectors: Time-to-Damage (TTD), pre-aim delta, and spray pattern deviation
- Spatial and Positional Features: Cross-match positioning consistency, Opening Duels context, and Trading Efficiency windows
- Tactical and Utility Metrics: Grenade/HE efficiency (Average HE damage), Clutch performance variance, and overall Kill/Death (KD)
- Social/Contextual Context: Premade lobby (Party) dependencies and cross-match behavioral

3.2.2 Data Sources and Profiling

- Synthetic telemetry generation: Simulating legitimate player profiles (varying from casual to highly-skilled) vs. simulated cheat profiles (aimbots, triggerbots, wallhacks and scripts)

3.2.3 Data Constraints and Limitations

- Limits of server-side logs
- Tick-rate limitations

3.3 Benchmark Execution and Raw Outputs

3.3.1 Experimental Protocol

- Setup of the controlled benchmarking environment
- Execution sequence: Batch processing of simulated profiles through the classical algorithm and the LSTM layer

3.3.2 Raw Evaluation Results

- Raw statistical outputs, distribution of suspicion scores and first results

3.4 Analysis and Performance Discussion

3.4.1 Net Detection Results

- Link between extreme data profiles and cheaters
- Robustness assessment: Evaluating the accuracy when mixing in high-skill profiles

3.4.2 Mitigating the High-Skill Bias

- Analyzing the conceptual limits of false positives
- Differentiating anomalous expert playstyles (exotic but legitimate behaviors) from software-assisted mechanics
- Comparative Analysis: Contrasting our suspicion-scoring pipeline with commercial performance platforms like Leetify.gg (How metrics designed for coaching can be repurposed for anti-cheat signaling)

3.4.3 Future Optimizations

- Build accurate data profiles faster
- Take into account external factors (number of cheaters in their friends list, age of the account, number of hours played related to the skill-level shown... things that experimented players learn to recognize but that the software fails to implement)

3.5 Comparison with Kernel-level approach

- Effectiveness
- Intrusiveness
- Implications for the FPS / e-sport industry
- Player perception
- Human validation

CONCLUSION

The easiest way to stop piracy is not by putting antipiracy technology to work. It's by giving those people a service that's better than what they're receiving from the pirates.

– GABE NEWELL

- Overall synthesis
- Limitations and future work
 - Lack of real data
 - Adaptation to further cheating methods

Ultimately, this thesis demonstrates that shifting toward server-side behavioral analysis resolves not only the critical security flaws of kernel-level access but also its severe socio-legal friction. By relocating the trust anchor from the user's local machine to the game server, developers can entirely bypass the ethical dilemmas associated with continuous Ring 0 surveillance. This paradigm shift aligns seamlessly with stringent global privacy regulations like the GDPR by eliminating the collection of non-game personal data. Furthermore, by replacing intrusive local monitoring with non-invasive gameplay analysis, this approach restores user trust and removes the barrier to adoption for privacy-conscious players, proving that safeguarding competitive fairness does not require compromising fundamental digital rights.

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GLOSSARY

A

Aimbot An automation cheat that systematically manipulates the view-angle variables stored within client process memory, enforcing automated frame-perfect vector corrections and target tracking that completely overrides the physical limitations of manual peripheral inputs. . . . **Page :** 39

C

Cheat Unsanctioned software, hardware, or system modifications designed to provide an asymmetric, unfair competitive advantage to a player by subverting authoritative game engine mechanics, spatial information visibility, or user-input processing pipelines. **Page :** 39

Client The local software instance of a video game running on the end-user's operating system, responsible for rendering local graphical vectors, capturing hardware peripheral inputs, and transmitting state packets asynchronously to a centralized authoritative network infrastructure. **Page :** 39

Counter-Strike A premier tactical round-based first-person shooter series developed by Valve, focusing on asymmetric objective match-ups between Terrorist and Counter-Terrorist factions, serving as the core empirical research and data evaluation framework for this thesis. . . **Page :** 39

Cybersecurity The scientific discipline and operational practice of protecting digital systems, networks, programs, and endpoints from unauthorized exploitation, access, or subversion, encapsulating the architectural evaluation of privileged kernel drivers versus server-side privacy boundaries. **Page :** 39

D

Driver A privileged kernel-mode software subsystem designed to abstract communications between the host operating system and physical hardware components, frequently weaponized by cheat developers via vulnerable third-party modules or deployed by anti-cheat systems for deep host visibility. **Page :** 39

E

E-Sport Electronic Sports, a structured global ecosystem of professional competitive video gaming organized into high-stakes leagues, where competitive integrity, technical fairness, and robust anti-cheat validation are paramount to financial and structural stability. **Page :** 39

F

First-Person Shooter A highly competitive genre of action video games centered on weapon-based combat rendered from a direct first-person graphical perspective, placing critical reliance on rapid mechanical precision, situational awareness, and split-second cognitive responses. . **Page :** 39

K

Kernel The foundational, core layer of an operating system operating at the highest execution privilege level (Ring 0), possessing unrestricted access to physical hardware channels, system volatile memory allocation, and low-level processor instructions. **Page :** 39

L

Leetify Leetify.gg, an external third-party data analytics platform that parses match-demo data streams to compute complex gameplay performance analytics (such as aim tracking and trading position efficiency), serving as the core conceptual inspiration for the server-side behavioral pipeline developed in this work. **Page :** 39

M

Microsoft Microsoft Corporation, the developer of the Windows operating system, whose secure boot architectures, device driver digital signature mandates, and low-level API design define the modern technical interface boundaries for client-side security software. **Page :** 39

O

Overwatch A decentralized crowdsourced tribunal system historically implemented within Valve's ecosystem, allowing experienced, trusted human reviewers to audit anonymized game replay telemetry to validate or reject suspicious flags produced by anti-cheat systems. . . . **Page :** 39

R

Radarhack An information cheat that extracts real-time physical spatial coordinates (X, Y, Z vectors) of hidden entities from volatile client memory space or intercepted network socket streams, projecting opponent positions onto an external clean secondary canvas or mini-map interface. **Page :** 39

Rings The hierarchical privilege rings architecture implemented within modern central processing units (CPUs) to isolate executing software, ranging from the highly restricted User Mode (Ring 3) to the absolute system authority of Kernel Mode (Ring 0). **Page :** 39

S

Script An execution sequence consisting of pre-compiled instructions or high-frequency input macros designed to automate complex, frame-dependent input timings, bypassing manual mechanical execution constraints such as perfect jump-throw synchronization or rapid spatial hops. **Page :** 39

Server The central, authoritative computing node in a multiplayer network architecture that validates game state logic, enforces simulation rules, manages synchronization across all connected clients, and generates deterministic telemetry data streams. **Page :** 39

T

Tick The elementary discrete discrete time-step execution unit of a multiplayer game server loop, defining the temporal frequency (measured in Hertz) at which the server processes client inputs, re-evaluates physical variables, and serializes state packets. **Page :** 39

Triggerbot An automation cheat that continuously polls internal entity lists or crosshair alignment parameters, instantly injecting synthetic mouse-click signals into the operating system input stream the exact microsecond an opponent's hit-box intersects with the player's crosshair coordinates. **Page :** 39

V

Valve Valve Corporation, the American video game developer and digital distribution publisher responsible for creating the Source and Source 2 game engines, the Steam ecosystem, and the authoritative competitive Counter-Strike series. **Page :** 39

W

Wallhack A category of information cheat that intercepts and alters client-side graphics rendering application programming interfaces (APIs) by disabling occlusion culling or depth-testing rules, granting an unauthorized real-time visual overlay of opponent models through solid opaque terrain geometry. **Page :** 39

ACRONYMS

API	Application Programming Interface	Page : 39
BYOVD	Bring Your Own Vulnerable Driver	Page : 39
CPU	Central Processing Unit	Page : 39
CS	Counter-Strike	Page : 39
CS2	Counter-Strike 2	Page : 39
CS:GO	Counter-Strike: Global Offensive	Page : 39
DKOM	Direct Kernel Object Manipulation	Page : 39
DMA	Direct Memory Access	Page : 39
ESP	Extra Sensory Perception	Page : 39
GRPD	General Data Protection Regulation	Page : 39
HE	High Explosive	Page : 39
LSTM	Long Short-Term Memory	Page : 39
OS	Operating System	Page : 39
PCIe	Peripheral Component Interconnect Express	Page : 39
RAM	Random Access Memory	Page : 39
TTD	Time To Damage	Page : 39

SOCIO-LEGAL, PRIVACY, AND COMMUNITY DYNAMICS OF KERNEL-LEVEL ANTI-CHEAT IMPLEMENTATIONS

A.1 System Visibility Level and Perceived Intrusion

The architectural integration of anti-cheat software into the kernel layer shifts the boundaries of data access. Operating at Ring 0, a driver possesses the same execution privileges as the core operating system, effectively rendering the traditional memory protection boundaries of user-mode (Ring 3) obsolete.

- **Theoretical Access to Non-Game-Related Data:** Because kernel drivers have unrestricted access to the entire Virtual Address Space, they can theoretically read and monitor memory pages belonging to unrelated applications. This includes open web browsers containing active sessions, password managers, encryption keys, and local files. While publishers assert that their drivers only scan for cheat signatures, the technical capability to perform deep packet and memory inspection across the entire system remains a valid point of concern for privacy advocates.
- **Continuous vs. Occasional Monitoring:** A major shift occurred with the introduction of Riot Games' Vanguard (vgk.sys), which mandates an "always-on" approach. Unlike traditional anti-cheats that initialize alongside the game client and terminate upon closure, always-on kernel drivers boot during the system startup sequence. The industry justification is to prevent the preloading of cheats or hypervisors before the anti-cheat can initialize. However, this creates a state of continuous monitoring, leaving a privileged third-party component active even when the user is performing sensitive tasks unrelated to gaming.
- **The Notion of Implicit Consent:** End-User License Agreements (EULAs) operate on a "take-it-or-leave-it" basis. Users are faced with an asymmetric choice: surrender Ring 0 access to their personal computer or lose access to the digital service entirely. In academic literature, this undermines the principle of freely given and informed consent, as the economic and social pressure to maintain connection with peer groups in modern video games forces users to accept invasive telemetric

conditions.

A.2 Social Perception and Community Acceptance

The deployment of kernel-level anti-cheats has transformed a purely technical challenge into a highly polarized cultural debate within the global gaming community.

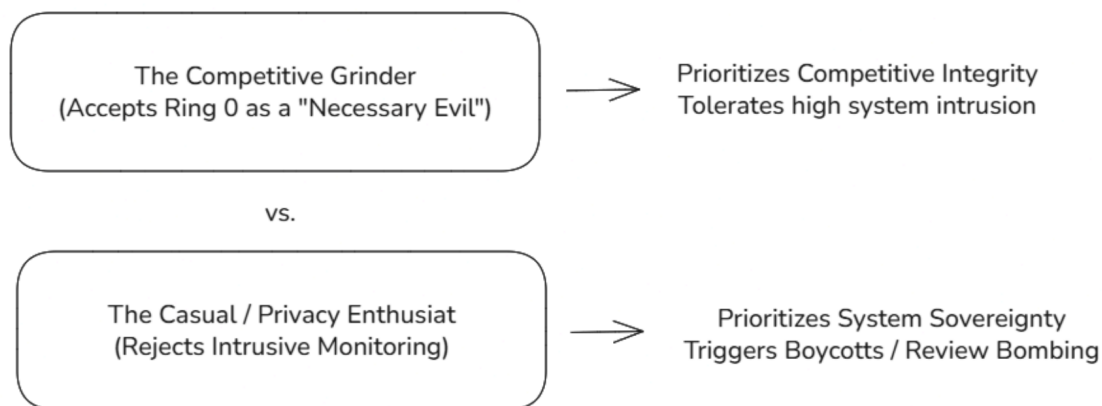


Figure 6: User Segmentation & Acceptance Divide, highlighting the contrasting perspectives of competitive and casual players regarding intrusive anti-cheat measures

- **Public Debates Around Intrusive Anti-Cheats:** High-profile releases have frequently triggered severe community backlash. For example, the launch of *Helldivers 2* faced significant criticism and review-bombing on platforms like Steam due to its implementation of nProtect GameGuard. Users frequently cite performance degradation, potential system instability, and the lack of transparency regarding what data is transmitted back to external servers as primary grievances.
- **The Divide Between Competitive and Casual Players:** The player base is structurally divided regarding acceptability, as summarized in Figure 6: competitive players tend to accept Ring 0 access as a necessary trade-off for integrity, while casual users and privacy advocates reject it as disproportionate, often responding with boycotts or review-bombing campaigns. Highly competitive players, particularly within esports ecosystems, often champion intrusive client-side measures, viewing them as a necessary tool to safeguard competitive integrity and financial stakes. Conversely, casual players or hardware enthusiasts view these measures as disproportionate. This divide is further amplified by ecosystem fragmentation, such as the complete exclusion of Linux and Steam Deck users from titles like *League of Legends* or *Apex Legends* due to the technical incompatibility of kernel drivers with translation layers like Proton.
- **Impact on Game Adoption:** This friction directly impacts the commercial viability of software. A growing segment of privacy-conscious consumers actively boycotts games utilizing kernel-level solutions. For publishers, this creates an operational trade-off: mitigating financial losses caused by cheaters versus losing potential customers who refuse to compromise their system sovereignty.

A.3 Legal and Regulatory Questions

From a legal perspective, the deep monitoring required by client-side kernel components tests the boundaries of modern data protection frameworks and civil liability.

- **GDPR Compliance: Proportionality and Data Minimization:** Under Article 5(1)(c) of the General Data Protection Regulation (GDPR), personal data processing must be adequate, relevant, and limited to what is necessary in relation to the purposes for which they are processed (data minimization). Proving that an anti-cheat requires sweeping visibility over a user's entire operating system to catch cheaters represents a precarious legal position. If a server-side or behavioral approach can achieve similar outcomes without local system surveillance, the kernel-level approach may fail the core GDPR test of proportionality.
- **Liability in Case of a Security Flaw:** Standard EULAs explicitly disclaim all corporate liability for software-induced damages, hardware conflicts, or data breaches. However, from a consumer rights perspective, the mandatory installation of a software component capable of creating an exploitable system vulnerability (e.g., the historical Capcom driver exploit or the Genshin Impact anti-cheat exploit used to deploy ransomware) challenges these liability waivers. Should a major anti-cheat driver be weaponized on a global scale, publishers could face unprecedented class-action lawsuits concerning negligence in secure software design.
- **Transparency and Terms of Service:** Most publishers offer limited visibility into the telemetry data collected by their kernel components. The lack of open-source auditing or independent third-party verification means users must blindly trust corporate assertions. To align with modern transparency mandates, anti-cheat developers face growing pressure to provide detailed documentation, open APIs, or clear, granular logs showing precisely what memory segments are being analyzed and when data is being exfiltrated to the cloud.

GAME CONTEXT AND RULES

B.1 Competitive Format

Counter-Strike 2 (CS2) is a tactical, round-based first-person shooter played in 5-versus-5 matches between a Terrorist (T) and a Counter-Terrorist (CT) side. In the default competitive mode, a match consists of up to 24 rounds (12 per half before sides switch), each lasting a maximum of approximately 1 minute 55 seconds. A round is won either by completing an objective (planting and detonating the bomb for the T side, or defusing it / eliminating the opposing team for the CT side), by eliminating the entire opposing team, or by the timer running out.

Each player starts a match with a fixed amount of in-game currency and earns or loses money based on round outcomes, kills, and objective completions. This currency is spent during a "buy phase" at the start of each round on weapons, grenades (utility), and armor (Kevlar vest and helmet). This economic layer means that a player's loadout — and therefore their potential for damage output — varies from round to round, which is directly relevant to the telemetry features discussed in Chapter 3 (e.g. "premade lobby dependencies", utility efficiency).

B.2 Core Mechanics Relevant to Cheating

Several mechanical systems are central to understanding why the cheats described in Chapter 1 provide such a significant advantage:

- Recoil and spray patterns: each weapon follows a fixed, deterministic recoil pattern when fired continuously ("spraying"). Skilled players memorize and compensate for this pattern by moving their mouse in the opposite direction ("spray control"). An aimbot or recoil-compensation script can replicate this perfectly and instantly, which is the mechanism illustrated in Figure 2.
- Movement and "counter-strafting": CS2 applies an accuracy penalty while a player is moving. Precise, frame-perfect counter-strafting (briefly tapping the opposite movement key to cancel momentum before shooting) restores full accuracy. This is a mechanical skill ceiling that automation cheats can trivially exceed.

- Sound and information design ("fog of war"): as introduced in Section 1.1.2, large parts of player decision-making rely on incomplete information — footstep sounds, gunfire direction, utility usage, and map knowledge. The "mini-map" only shows teammates and recently-spotted enemies. This deliberate information asymmetry is the foundation that wallhacks and radarhacks subvert.
- Damage model and hit zones: unlike some shooters, CS2 does not apply a fixed "headshot multiplier" as a flat bonus; rather, each of the four hit zones on a player model (head, chest/arms, stomach, legs) carries its own damage multiplier applied to the weapon's base damage, with the head dealing roughly four times the damage of a torso hit, the stomach somewhat more than the torso, and the legs somewhat less. This zone-dependent damage model is precisely why aim assistance (aimbots) is so impactful: it does not merely guarantee a hit, it guarantees a hit on the highest-value zone (the head), turning encounters that would normally require multiple shots into immediate eliminations.

WHY CHEATING BREAKS GAME BALANCE: THE BULLET PENETRATION CASE

C.1 The Penetration Power Mechanic

In CS2, every weapon has a statistic generally referred to as "Penetration Power" (sometimes informally called "obstacle penetration"), which is distinct from "Armor Penetration" (the percentage of damage retained against a Kevlar vest, discussed separately in commercial weapon guides). Penetration Power governs how much damage a bullet retains after passing through an environmental surface — a wall, door, crate, or similar obstacle.

Crucially, this statistic is not expressed as a simple percentage and is not uniform across weapon categories. As a general rule:

- Pistols and submachine guns have the lowest penetration values and lose most of their damage when fired through cover, often dealing negligible or zero damage to a target behind a moderately thick wall.
- Assault rifles (AK-47, M4A4/M4A1-S, etc.) have a markedly higher penetration value, allowing them to retain a meaningful fraction of their damage through thin-to-medium obstacles such as wood, drywall, or thin metal.
- Sniper rifles (AWP, SSG 08, SCAR-20, G3SG1) have the highest penetration values in the game, making them the most effective weapons for shooting through cover.

The actual damage delivered after penetration depends on three combined factors: the weapon's Penetration Power, the material and thickness of the obstacle (thin drywall absorbs little damage; concrete or steel absorbs much more or blocks the shot entirely with the total number of surfaces a single bullet can penetrate capped at four), and the distance the bullet has already travelled before and after penetration with thin metal and drywall retaining 40–70% of damage while concrete or multi-layered walls can reduce damage by 70–90%, often rendering low-penetration weapons such as SMGs or pistols nearly useless.

C.2 Link to Information Cheats

This mechanic is the missing piece that explains why wallhacks (Section 1.1.2) represent such a disproportionate advantage, beyond simply "seeing through walls":

1. Legitimate uncertainty: an honest player knows, in principle, that bullets can pass through certain surfaces, and experienced players memorize a limited number of well-known "wallbang" spots on each map. However, they cannot know, in real time, the exact position of an unseen enemy behind that surface, nor precisely how much damage their shot will deal once combined factors (material, thickness, distance, weapon) are taken into account. Firing through a wall "blind" is therefore a probabilistic, high-risk action — a guess informed by sound cues, game sense, and map knowledge, not a guaranteed kill.
2. What a wallhack removes: a wallhack (Figure 3) eliminates this uncertainty entirely. The cheater sees the enemy's exact silhouette, position, and hit zone (head, chest, etc.) through the wall in real time. Combined with the penetration mechanic above, this transforms a probabilistic "blind" shot into a deterministic one: the cheater can wait until the visible target's head is aligned with a high-penetration weapon (e.g. an AWP or rifle) and a sufficiently thin surface, then fire with full confidence of landing a lethal headshot — something a legitimate player could never reliably replicate without the visual information.
3. Compounding effect with aim assistance: when combined with an aimbot, the penetration mechanic becomes almost irrelevant as a defensive layer for legitimate players, since the cheat can both locate the target through any surface and snap the crosshair to the highest-damage hit zone the instant a penetrable line of sight exists.

This illustrates concretely why the "fog of war" described in Section 1.1.2 is not a cosmetic feature but a core balancing mechanism: the penetration system was designed by Valve under the assumption that players cannot see what they are shooting at, and that the resulting uncertainty offsets the raw power of high-penetration weapons. Information cheats break this assumption without requiring any change to the weapon's underlying damage values, which is also why such cheats are extremely difficult to detect from server-side damage logs alone — the shot itself is mechanically "legal."

¹https://counterstrike.fandom.com/wiki/Bullet_Penetration

²<https://cs2pulse.com/wallbangs/>

CLIENT/SERVER ARCHITECTURE AND NETWORK COMMUNICATION

D.1 The Authoritative Server Model

CS2 follows a client-server architecture in which the server is authoritative over the game state. At a fixed frequency (the "tick rate"), the server receives input commands from each connected client (movement, view angles, button presses), simulates the resulting game state (player positions, weapon fire, damage, bomb state, etc.), and serializes a snapshot of the world state back to each client.

D.2 Snapshot Scope and Client Memory

Each client receives a snapshot covering all entities in the game world — including the positions, health values, and state of every player, regardless of their visibility to the receiving client. The client-side engine is responsible for rendering only what the local player should legitimately see; the underlying data for all entities is present in the client process's memory regardless.

After reception, the snapshot is decrypted and parsed into the Entity List — a data structure held in RAM containing plaintext 3D coordinates, health values, velocity vectors, and other attributes for every entity in the game. This in-memory representation is what the local engine uses to run prediction and rendering logic.

D.3 Network Channel Encryption

CS2 traffic is relayed through Valve's Steam Datagram Relay (SDR) infrastructure. The underlying networking library performs a Diffie-Hellman key exchange to establish an encrypted channel between client and server. Traffic is additionally relayed so that neither players' nor servers' IP addresses are exposed to each other. As a result, the snapshot payload is encrypted end-to-end at the transport layer.

D.4 Memory-Based Exploit Surface

Because the full world state is present in plaintext in the client process's RAM after decryption, it can be read by any component with access to that memory space. Radarhacks and wallhacks operate on this basis: a separate process reads the Entity List and uses the extracted coordinates to render enemy positions on a minimap or through geometry — data the client holds locally but would not normally display.

D.5 DMA Cheats

Direct Memory Access (DMA) cheats exploit the same condition via a hardware peripheral, typically connected via PCIe, that reads host memory from outside the operating system's process model. Because the access occurs at the hardware level, it does not appear as a software memory operation within the OS, making it transparent to both userland and kernel-level monitoring.

OTHER APPENDIXES

E.1 Cheaters in CS2

Community-driven data aggregation suggests that the actual prevalence of cheaters in CS2 matchmaking is lower than subjective perception implies — Figure 7 shows one such analysis, estimating a 6.6% chance of a cheater being present in a given game across approximately 3,000 matches at an average elo of 6,900.



Figure 7: Data aggregation made by a CS2 player to show the chance of a cheater being in a game

E.2 Existing data aggregation tool

Third-party platforms already aggregate match telemetry into rich per-player behavioral profiles, as shown in Figure 8, demonstrating that demo-derived data is sufficiently granular to serve as a foundation for the suspicion-scoring pipeline developed in Chapter 3.



Figure 8: Existing data aggregation tool to help visualize a player's game statistics

²<https://csst.at/profile/76561198261637028>

Abstract: Awesome abstract. [1, 2, 3, 4, 5] Cheat Client Server Kernel Valve Wallhack Radarhack Aimbot Triggerbot Script First-Person Shooter Counter-Strike E-Sport Cybersecurity Rings Microsoft Driver Overwatch Tick Leetify CS CS2 CS:GO OS DMA GRPD API ESP BYOVD DKOM PCIe RAM CPU LSTM TTD HE

Keywords: Key, Words.